

Evaporation

Final Lab Report 1

Unit Operations

Lab 1

February 3, 2020

Group 1

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1. Executive Summary (10 pts)

The goal of this experiment was to concentrate a solution by single-effect evaporation and observe the effects that three pressures have on the heat transfer rate. A 10 wt % sugar solution was utilized for the system at the following pressures: 6, 12 and 19 inHg. The results showed that the capacity and economy of the system were directly proportional to increasing vacuum pressure. The capacities were 0.128, 0.194, and 0.244 and economies 0.219178, 0.312903, and 0.357771 in ascending vacuum pressure. Increase in vacuum pressure also increased the temperature difference of the system, which yielded an increased heat transfer rate.

In conclusion, the experiment succeeded as the results were consistent with the theory expectations. The capacity and economy remained proportional as the vacuum pressure was changed. Furthermore, the single-effect evaporation is not cost and energy efficient, and a multi-effect system is recommended because the energy leaving the single-effect system is discarded, thus increasing steam costs. It is more ideal to use single-effect evaporation when the required capacity is relatively small.

2. Introduction (10 pts)

The experimental objective was to determine the capacity and economy at 3 different vacuum pressures using a single-effect evaporator at a constant concentration. The single effect evaporator is composed of the evaporator (heat exchanger), the condenser, the separator, the vacuum pump and a valve to add makeup water and to drain the vessel.

The significance of this experiment was to become familiar with the single-effect evaporation process by vaporizing a portion of the solvent to produce a concentrated solution or thick liquor [1]. Evaporators are most commonly used in processing industries such as food, dairy, dyes, paints and pigments [2]. Evaporators are also used in waste management and for low temperature applications in food and pharmaceutical industries [2].

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3. Theory (10 pts)

In a steam -heated tubular evaporator, the two principal measures of performance are the capacity and the economy [3]. Capacity is defined as kg of water vaporized/hr. Economy is defined as kg vaporized/kg of steam fed whereas, the economy is almost always less than 1 in a single effect evaporator.

The evaporator capacity is affected by changes in the temperature drop and the overall heat transfer coefficient. If the evaporator feed is at boiling temperature, all the heat transferred through the heating surface can be evaporated and the capacity is proportional to the heat transfer rate. The heat transfer rate is defined as:

$$q = UA\Delta T$$

Where:

U is the overall heat transfer coefficient

A is the heat transfer surface

ΔT is the overall temperature drop

The temperature drop is fixed by the properties of the steam and the boiling liquid and the overall heat transfer coefficient is influenced by the design and mode of operation of the evaporator [3].

Although, the evaporator economy is affected by the number of stages and the temperature of the feed, the economy is entirely a matter of enthalpy balances. For single effect evaporation, the enthalpy balances for steam and the vapor or liquid side is needed [3]. The enthalpy balance for the steam side is defined as:

$$q_s = \dot{m}_s(H_s - H_c) = \dot{m}_s\lambda_s$$

Where:

q_s is rate of heat transfer through heating surface from steam

H_s is the specific enthalpy of steam

H_c is the specific enthalpy of condensate

λ_s is the latent heat of condensation of steam

$\dot{m}_s$ is the rate of flow of steam

The enthalpy balance for the liquor side is defined as:

$$q = (\dot{m}_f - \dot{m})H_v - \dot{m}_f H_f + \dot{m}H$$

Where:

q is the rate of heat transfer from heating surface to liquid

H_v is the specific enthalpy of vapor

H_f is the specific enthalpy of thin liquor

H is the enthalpy for thick liquor

$\dot{m}_f$ and $\dot{m}$ is the rate of steam flow of the thin liquor or feed and the rate of steam flow of the thick liquor respectively

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Show the exact equation for the capacity and the economy.

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4. Results (15 pts)

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Based on **Table 3**, the boiling point of the solution at different pressures can be determined and the capacity, economy and steam consumption can be calculated for all three runs; sample calculations are shown in **Appendix III**. The resulting values are shown in **Table 1** and several trends can be observed in this experiment. As the vacuum pressure increases both the capacity and economy increase as shown in **Figure 1** and **Figure 2** respectively; the economy of the evaporator gets closer to unity as the vacuum pressure increases. Also, as the vacuum pressure increases, the solution's boiling point decreases which is measured by Thermocouple 1.

There are also several trends that are noticeable in the temperature changes as shown in **Table 2**. One assumption that was made is that the room temperature was 25°C; this value was used to calculate the change in temperature of the solution through the heat exchanger. As the vacuum pressure increases, the temperature change of the steam in the heat exchanger, the temperature change of the steam in the condenser and the temperature difference between the steam and solution increase. Inversely, as the vacuum pressure increases, the temperature change of the sugar solution in the heat exchanger, the temperature change of the sugar solution in the condenser and the temperature difference of the cooling water through the condenser decrease. The energy from the heat exchanger increases as the vacuum pressure increase as seen in **Figure 3**.

Table 1: Calculated Results

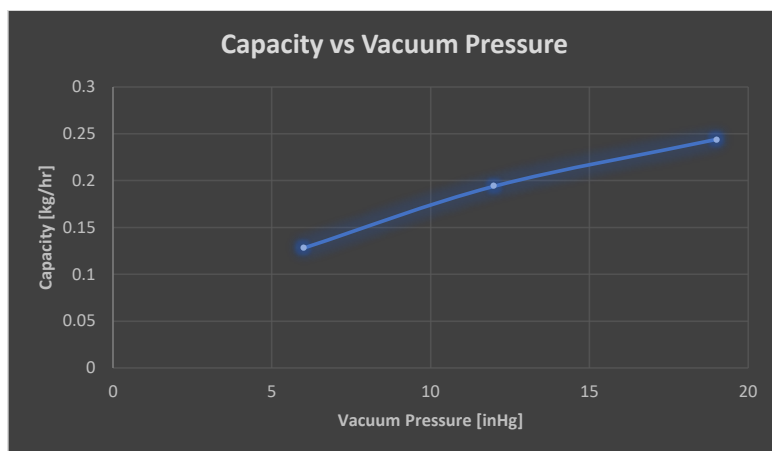
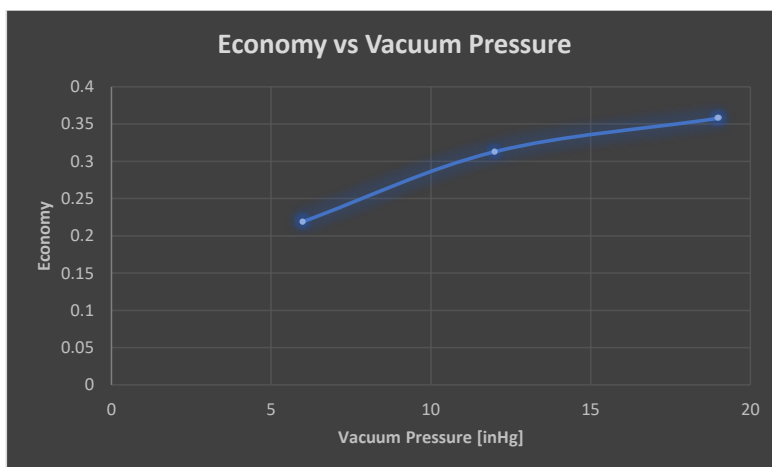
	Run 1	Run 2	Run 3
Vacuum Pressure (inHg)	12	19	6
Mass, condensate (kg)	0.0485	0.061	0.032
Mass, vapor (kg)	0.155	0.1705	0.146
Time (hr)	0.25	0.25	0.25
Capacity (kg/hr)	0.194	0.244	0.128
Economy	0.312903	0.357771	0.219178
Steam Consumption (kg/hr)	0.62	0.682	0.584

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Table 2: Temperature Changes for Different Vacuum Pressures

	Run 1	Run 2	Run 3
Vacuum Pressure (inHg)	12	19	6
ΔT (steam through HeatX)	-10°C	-13°C	-9°C
ΔT (steam through condenser)	-68°C	-74°C	-66°C
ΔT (solution through HeatX)	60°C	46°C	67°C
ΔT (solution through condenser)	-76°C	-61°C	-85°C
ΔT (steam - solution){HeatX outlet}	18°C	29°C	12°C
ΔT (cooling water through condenser)	21°C	16°C	24°C

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**Figure 1:** Capacity vs Vacuum Pressure**Figure 2:** Economy vs Vacuum Pressure

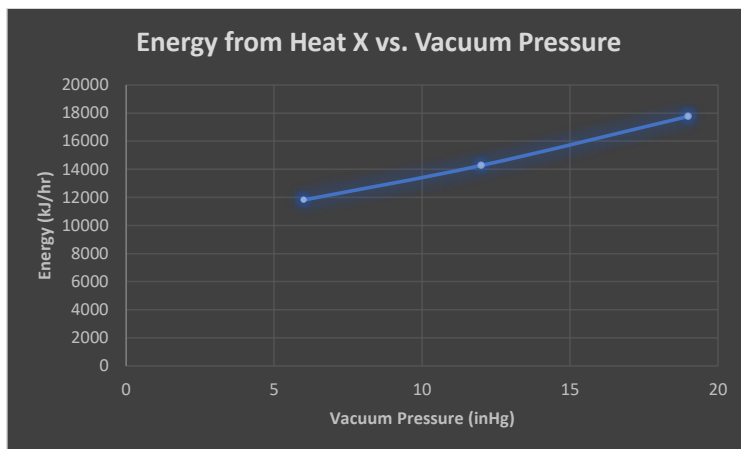


Figure 3: Energy from the Heat Exchanger vs Vacuum Pressure

5. Discussion (15 pts)

From the results, it is evident that the theory is supported by the results. As the vacuum pressure is increased, the boiling point of the solution is decreased. When the boiling point of the solution decreases, less steam is required to overcome the latent heat of the water in the solution per unit volume. Furthermore, you have an increase in the amount of water in solution vaporized. This follows the trend that the economy and the capacity both increase with increased vacuum pressure. There is more water being vaporized, and less steam required overcome that vaporization.

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The heat of vaporization does change at different pressures. This is made evident by the amount of steam condensate recovered after the steam trap. At higher vacuum pressures, the boiling point temperature of the water decreased. At lower boiling point temperatures, the heat of vaporization is higher.

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The economy and capacity of the process could be improved with an increase in the number of stages in the experimental apparatus or by the inclusion of a second evaporator. It might be beneficial for improving the process to have a system with more stages or effects available to be able to accurately determine which specification of the system affects which results. Having the ability to determine the relationship between the amount of stages and the economy would be helpful in other applications.

There was however an issue with the readings that were taken after the 15-minute waiting window. This is explained in further detail in Appendix II. The values that were taken at the different pressures due to the moving of the valve are still consistent with the theory and are supported by the results. Even though the pressures should not have changed in that 15-minute window, the vibration from the pump did make the valve close a little bit and therefore the values that were recorded, reflected that.

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6. References (10 pts)

- [1] Zdunek, Alan. (Fall 2014). "Evaporation." ChE 381: Unit Operations Laboratory. University of Illinois at Chicago Chemical Engineering Department
- [2] "Common Uses and Applications for Evaporators." Thermal Kinetics. (2018). Retrieved from: <https://thermalkinetics.net/common-uses-applications-evaporators>
- [3] McCabe W.L., Smith J.C., Harriott P. "Unit Operations of Chemical Engineering", Chapter 16, 7th Edition. McGraw-Hill, NY, NY 2005
- [4] Physical Properties of Water
Retrieved from: https://www.engineeringtoolbox.com/water-properties-d_1573.html

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7. Appendix I: Data Tabulation/Graphs (10 pts)

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Table 3: Data at Three Vacuum Pressures

	Run 1	Run 2	Run 3
Flowrate of cooling water (gal/min)	0.38	0.38	0.38
Vacuum Pressure (inHg)	12	19	6
Level of steam condensate (mL)	48.5	61	32
Level of collected vapor (mL)	155	170.5	146
Thermocouples (deg C)			
Water Vapor (out from HeatX), 1	85	71	92
Water Vapor (Inlet to Condenser), 2	84	70	91
Condensed Water Vapor (Out from Condenser), 3	8	9	6
Steam out HeatX (inlet to Steam Trap), 4	103	100	104
Cooling Water (outlet from Steam trap), 5	27	21	28
Cooling Water (inlet from steam trap), 6	6	5	4
Condensed Steam (out from steam trap), 7	35	26	38
Inlet Steam, 8	113	113	113

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8. Appendix II: Error Analysis (10 pts)

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Quantitative analysis is needed. What if the solution level varies ~5%? Try to make error bars according to this.

There are several parameters that could have caused errors during the experiment. One of them is that the pressure was not constant throughout each run. This was due to the fact that the valve tends to vibrate and loosen; thus, it should have been secured in order to maintain a constant vacuum pressure, which we did not do. Another factor that could have affected the data is that the make-up water that was introduced into the system was at room temperature when it mixed with the sugar solution before was fed into the heat exchanger. The final source of error is that the sugar solution in the separator vessel had to be kept at a constant level. However, this was difficult to do because of the vigorous flowrate that took place and that the level was visually estimated.

9. Appendix III: Sample Calculations (10 pts)

Mass = density of water * level of steam or vapor

$$\mathbf{0.0485\ kg} = 48.5\ \text{mL} * 0.001\ \text{kg/mL}$$

Capacity = mass of steam condensate/hr

$$\mathbf{0.194\ kg/hr} = 0.0485\ \text{kg} / 0.25\ \text{hr}$$

Economy = mass vaporized / mass of steam fed

$$\mathbf{0.3129} = 0.0485\ \text{kg} / 0.155\ \text{kg}$$

Steam consumption = capacity / economy

$$\mathbf{0.62\ kg/hr} = 0.194\ \left(\frac{\text{kg}}{\text{hr}}\right) / 0.3129$$

$$q_{\text{coolingwater}} = \dot{m}C_p\Delta T$$

$$q_{\text{coolingwater}} = \left(86.30712\frac{\text{kg}}{\text{hr}}\right)\left(4.187\frac{\text{kJ}}{\text{kg} * \text{K}}\right)(300 - 279\ \text{K})$$

$$q_{\text{coolingwater}} = \mathbf{7,588.73\frac{kJ}{hr}}$$

$$q_{\text{total}} = \dot{m}_{\text{steam}}C_p\Delta T + \dot{m}_{\text{steam}}\Delta H_{\text{vap}}$$

$$q_{\text{HEX}} = q_{\text{total}} - q_{\text{coolingwater}}$$

$$q_{\text{HEX}} = \mathbf{14271.81\ kJ/hr}$$

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Specify the calculation you presented.

10. Appendix IV: Individual Team Contributions

NAME: Sandra Hernandez

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	2	
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)	3	
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	2	
Error Analysis (final lab)		
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	18	

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NAME: Tianna Mitchell

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)		
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)		
Introduction (Final lab)	1	
Theory (Final Lab)	1	
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	2	
Error Analysis (final lab)		
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	18	

Evaporation**University of Illinois at Chicago**

NAME: Anne Serban

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)	1	
Materials & Supplies (prelab)		
Procedure (prelab)	1.5	
Project Safety Evaluation (prelab)		
Final Lab Editing	1.5	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	2	
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	1.5	
Error Analysis (final lab)		
Sample Calculations (final Lab)	1.5	
Power Point Presentation	1	
Total	18	

Evaporation**University of Illinois at Chicago**

NAME: Maurice Smith

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)	1.5	
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	1.5	
Final Lab Editing	1.5	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	2	
Conclusion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	1.5	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	18	